SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

ICT 133
Structured Programming

Tutor-Marked Assignment

January 2025 Semester Presentation

TUTOR-MARKED ASSIGNMENT (TMA)

This assignment is worth 24" of the final mark for ICT 133, Structured Programming.

The cut-off date for this assignment is **Sunday, 16 March 2025, 2355hrs.**

Note to Students:

All TMAs must be submitted electronically through Canvas. No partial submission of TMA will be accepted unless otherwise specified.

Please refer to undergraduate student handbook for more information:
Combined UG Student Handbook January 2025 (suss.edu.sg)

Assignment Requirements:

- Do **NOT** define classes for this TMA.
- **Unless specified in the question, you CANNOT use packages not covered in this module, e.g., re, collections, numpy, pandas etc.**
- All functions must be documented. Provide sufficient comments to your code and ensure that your program adheres to good programming practices such as not using global variables.
- **Do not use exit() function.**
- Failing to do so can incur a penalty of as much as 50% of the marks allotted.

Submission Details:

- Use the template word document provided - **SUSS_PI_No-FullName_133TMA.docx.** Rename the file with your SUSS PI and full name join with "_TMA" e.g., "**SUSS_PI-TomTanKinMeng_133TMA.docx**" (without the quotes).
- Include the following particulars on the first page of the word document, on separate lines: **Course Code, SUSS PI No., Your Name, Tutorial Group and Submission Date.**
- Copy and paste the source code of each program you write in **text** format. Submit screenshots for **only** output of your program, where applicable.
- If you submit the source code as a screenshot, your code will not be marked. That is, **screenshot code will be awarded zero mark.**
- Submit your solution in the form of a **single MS Word document. Do NOT submit as a pdf document.** You will be penalised if you fail to submit a word document.
- The word document must be submitted to your respective T group. Besides this, you are required to upload your source code to Vocareum.

Question 1 (20 marks)

This question covers materials in Seminar 1 and 2. Use and express selection structure for this question. Functions, repetition, or collections are not necessary for this question.

As part of the Assurance Package (AP), every Singaporean aged 21 years and above, will receive AP Cash payments in December every year from 2022 to 2026, the amount depending on his/her income and property ownership.

Singaporeans aged 21 years and above in reference year	Disbursement Period	Owns 0 to 1 property			Owns more than 1 property
		Assessable Income (AI)			
		Up to $34,000	More than $34,000, and up to $100,000	More than $100,000	
2023	Dec 2022	$200	$150	$100	$100
2024	Dec 2023	$600	$350	$200	$200
2025	Dec 2024	$600	$350	$200	$200
2026	Dec 2025	$600	$350	$100	$100
2027	Dec 2026	$250	$150	$100	$100
	Total	**$2,250**	**$1,350**	**$700**	**$700**

Table 1: AP Cash Quantum

a) Write a program that helps Singaporeans find out the estimated AP Cash amount they will receive from the Government in 2026. Your program will ask a user to input ONLY the relevant parameters necessary to estimate this amount. Relevant parameters include birth year, number of properties owned, and assessable income. For example, if the user is younger than 21 years, it is not necessary to ask for the rest as he/she is not eligible for AP Cash. Assume the user will enter valid inputs to the various parameters. Based on Table 1, your program shall determine and display the eligible amount of AP Cash (if qualify). Submit and paste screenshots of at least FOUR program executions, with different inputs.

Sample 1:
```
Enter year of birth:2006
You are only 20 in 2026, not eligible for AP Cash
```

Sample 2:
```
Enter year of birth:2005
Do you own more than 1 property? (Y/N):N
Assessable Income:30000
You will receive $600 AP Cash in Dec 2025
```

Sample 3:
```
Enter year of birth:1970
```

```
Do you own more than 1 property? (Y/N):y
You will receive $100 AP Cash in Dec 2025
```

Sample 4:

```
Enter year of birth:1965
Do you own more than 1 property? (Y/N):n
Assessable Income:80000
You will receive $350 AP Cash in Dec 2025
```

(10 marks)

b) Write another program that will estimate the total amount a user will receive from the Government over all 5 years. If the user turns 21 between 2024 to 2027 (inclusive), assume that he/she does not own property and his assessable income is less than $34,000. In this version, instead of asking the user if he/she owns multiple properties, ask him/her from which reference year he/she started owning multiple properties and apply Table 1 accordingly. Submit and paste screenshots of at least FOUR program executions, with different inputs.

Sample 1:

```
Enter year of birth:2005
You will receive a total of $850 AP Cash over five years.
```

Sample 2:

```
Enter year of birth:2002
From which reference year do you own >1 property (0 for NA):0
Assessable Income:34000
You will receive a total of $2250 AP Cash over five years.
```

Sample 3:

```
Enter year of birth:1970
From which reference year do you own >1 property (0 for NA):2025
Assessable Income:80000
You will receive a total of $900 AP Cash over five years.
```

Sample 4:

```
Enter year of birth:1965
From which reference year do you own >1 property (0 for NA):2000
You will receive a total of $700 AP Cash over five years.
```

(10 marks)

Question 2 (30 marks)

This question covers materials up to Seminar 3. Make use of functions, selection and repetition structures. NO data structures like sets, lists or dictionary should be used for this question. The eval() function should not be used. Keep the program modular by defining other functions if necessary.

This program allows a user to practice addition, subtraction and multiplication of numbers. The requirements are as follows:

1. Allow user to input the highest level to complete. Users start from level 1 and the highest possible level is 5.

2. The question asked can be addition, subtraction or multiplication of 2 numbers. The type of operator is randomly chosen.
3. Addition. For level 1, both operands consist of single digits 1 to 9. For level 2, both operands are double digits and for level 3, both operands are 3 digits and so on. All the numbers should be randomly generated.
4. Subtraction. The number of digits for the operands for each level are the same as addition, except that the second operand must be less than the first, so that the result of subtraction will always be strictly greater than 0.
5. Multiplication is the same as addition in terms of the number of digits for the first operand at each level. For level 1, the first operand should not start with 1. The second operand will always be a single digit from 2 to 9 for all levels.
6. The exercise starts at level 1. If the user answers the question correctly, the user will proceed to the next level if he/she is not already at the highest level. If the user is already at the highest level, the program will exit with the message "You have passed."
7. If the user gave an incorrect answer, the user would drop down one level if he/she is not already at level 1. If the user is already at level 1, the program will exit with the message "You have failed."

a) Write a function `getExpression(level, operator)` which has one integer parameter representing a level and another string parameter representing the arithmetic operator. The function generates a string expression as well as the answer, based on the requirements for the level. Both the string expression and the result value are returned.

For example:

- `getExpression(1, "+")` may returns `"3 + 1 = ", 4`
- `getExpression(2, "-")` may returns `"23 – 17 = ", 6`

(10 marks)

b) Apply structured programming and implement the program. Include program execution screenshots of at least TWO different scenarios.

<u>Sample 1:</u>

```
Enter highest level to complete (1-5):6
Highest level must be between 1 and 5.
Enter highest level to complete (1-5):0
Highest level must be between 1 and 5.
Enter highest level to complete (1-5):3
Level 1, 2 + 9 = ? 11
Correct! Well done! Proceeding to the next level 2.
Level 2, 51 - 19 = ? 32
Correct! Well done! Proceeding to the next level 3.
Level 3, 837 * 2 = ? 1674
Correct! Well done! You have completed the highest level 3! You have
passed.
```

<u>Sample 2:</u>

```
Enter highest level to complete (1-5):3
Level 1, 2 + 3 = ? 5
Correct! Well done! Proceeding to the next level 2.
```

```
Level 2, 23 * 7 = ? 161
Correct! Well done! Proceeding to the next level 3.
Level 3, 644 + 322 = ? 1
Wrong! The answer is 966. Sorry, retake level 2 again.
Level 2, 52 - 40 = ? 12
Correct! Well done! Proceeding to the next level 3.
Level 3, 558 + 672 = ? 1
Wrong! The answer is 1230. Sorry, retake level 2 again.
Level 2, 74 * 6 = ? 1
Wrong! The answer is 444. Sorry, retake level 1 again.
Level 1, 2 - 1 = ? 1
Correct! Well done! Proceeding to the next level 2.
Level 2, 91 * 5 = ? 1
Wrong! The answer is 455. Sorry, retake level 1 again.
Level 1, 9 * 5 = ? 1
Wrong! The answer is 45. Sorry, you have answered wrongly at level 1!
You have failed.
```

(10 marks)

c) Enhance the program in (b) to give users more practice at each level before proceeding to the next. At the start, ask the user to input the number of consecutive correct answers N required before proceeding to the next level. Replace requirements 6 & 7 above with the new 6’ and 7’ below. Include program execution screenshots of at least TWO different scenarios.

6'. The exercise starts at level 1. If the user answers the question correctly, another question of the same level will be asked. With N consecutively correct answers, the user will proceed to the next level if he/she is not already at the highest level. If the user is already at the highest level, the program will exit with the message ‘You have passed.’

7'. If the user gave an incorrect answer, another question of the same level will be asked. If N incorrect answers are given consecutively, the user will drop down one level. If the user is already at level 1, the program will exit with the message ‘You have failed.’

<u>Sample 1:</u>

```
Enter number of correct answers before proceeding to the next level
(2-5):2
Enter highest level to complete (1-5):4
Level 1, 6 * 8 = ? 48
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 1, 8 * 2 = ? 16
Correct! Well done! You have answered 2 questions correctly,
proceeding to the next level 2.
Level 2, 16 * 3 = ? 48
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 2, 82 * 6 = ? 1
Wrong! The answer is 492. If you answer the next 1 questions wrongly,
you will drop one level.
Level 2, 96 * 7 = ? 672
```

```
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 2, 98 + 71 = ? 169
Correct! Well done! You have answered 2 questions correctly,
proceeding to the next level 3.
Level 3, 551 + 137 = ? 688
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 3, 532 - 349 = ? 183
Correct! Well done! You have answered 2 questions correctly,
proceeding to the next level 4.
Level 4, 9682 * 6 = ? 58092
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 4, 2868 - 1399 = ? 1469
Correct! Well done! You have completed the highest level 4! You have
passed.
```

Sample 2:

```
Enter number of correct answers before proceeding to the next level
(2-5):3
Enter highest level to complete (1-5):3
Level 1, 4 - 1 = ? 3
Correct! Answer the next 2 questions correctly to proceed to the next
level.
Level 1, 4 + 5 = ? 9
Correct! Answer the next 1 questions correctly to proceed to the next
level.
Level 1, 6 * 5 = ? 30
Correct! Well done! You have answered 3 questions correctly,
proceeding to the next level 2.
Level 2, 77 + 51 = ? 128
Correct! Answer the next 2 questions correctly to proceed to the next
level.
Level 2, 77 - 58 = ? 1
Wrong! The answer is 19. If you answer the next 2 questions wrongly,
you will drop one level.
Level 2, 49 * 2 = ? 1
Wrong! The answer is 98. If you answer the next 1 questions wrongly,
you will drop one level.
Level 2, 68 + 99 = ? 1
Wrong! The answer is 167. You have answered 3 questions wrongly,
retake level 1 again.
Level 1, 3 * 8 = ? 1
Wrong! The answer is 24. If you answer the next 2 questions wrongly,
you will drop one level.
Level 1, 3 * 4 = ? 1
Wrong! The answer is 12. If you answer the next 1 questions wrongly,
you will drop one level.
Level 1, 9 + 5 = ? 1
Wrong! The answer is 14. Sorry, you have answered 3 questions wrongly
at level 1! You have failed.
```

(10 marks)

Question 3 (25 marks)

This question covers materials up to Seminar 4. Employ structure programming and use of functions to make the program modular.

Singapore NRIC numbers start with "S" or "T" followed by 7 digits and ends with a reference letter which is determined by the following algorithm.

1. Multiply each digit by its corresponding weight (2, 7, 6, 5, 4, 3, 2), first digit by 2, second by 7 and so on
2. Sum the above products
3. If the NRIC starts with "T", add 4 to the sum
4. Find the remainder when the sum is divided by 11
5. Take 11 – remainder to get the check digit
6. Lookup the reference letter in this string "ABCDEFGHIZJ" where check digit 1 is A, 2 is B and so on

For example, the NRIC number "S7928964" will end with "G"

i. Sum = 7*2 + 9*7 + 2*6 + 8*5 + 9*4 + 6*3 + 4*2 = 191
ii. Remainder = 191%11 = 4
iii. Check digit = 11 – 4 = 7
iv. Reference letter = "G"

a) `verifyNRIC(nric: str) -> bool:`

Write the above function which has a string parameter representing the NRIC number. The function checks whether the reference letter is correct. The function shall return True if it is a valid NRIC number, False otherwise.

<u>Sample:</u>
```
>>> verifyNRIC('S7928964G')
True
```

(5 marks)

b) `guessNRIC(masked: str) -> list`

Write the above function which has a string parameter representing a NRIC number with 1 to 4 of its digits masked with the letter "X", e.g. "S79XX964G". The function shall print an error message if the parameter does not satisfy this format. The function shall return a list of all valid NRIC numbers with different permutations of the masked digits, e.g. ['S7906964G', 'S7917964G', 'S7928964G', 'S7939964G', 'S7950964G', 'S7961964G', 'S7972964G', 'S7983964G', 'S7994964G']. The function shall not use recursive calls and does not include nesting of more than 3 levels of loops

Using the above function, write the `main()` function to ask a user to enter a masked NRIC and print out the list of all valid permutations of the masked NRIC number. Loop continuously until the user enters an empty string.

<u>Sample:</u>
```
Enter NRIC with 1-4 masked digits (<Enter> to exit):f79xx264g
Invalid NRIC! NRIC should start with S or T

Enter NRIC with 1-4 masked digits (<Enter> to exit):s79xxxxxg
```

```
Number of masked digits should be between 1 to 4.

Enter NRIC with 1-4 masked digits (<Enter> to exit):s79x896xg
s79x896xg has 9 permutations: ['s7918967g', 's7928964g', 's7938961g',
's7948969g', 's7958966g', 's7968963g', 's7978960g', 's7988968g',
's7998965g']

Enter NRIC with 1-4 masked digits (<Enter> to exit):s79xx96xg
s79xx96xg has 91 permutations: ['s7900968g', 's7901960g',
's7902963g', 's7903966g', 's7904969g', 's7905961g', 's7906964g',
's7907967g', 's7909962g', 's7910965g', 's7911968g', 's7912960g',
's7913963g', 's7914966g', 's7915969g', 's7916961g', 's7917964g',
's7918967g', 's7920962g', 's7921965g', 's7922968g', 's7923960g',
's7924963g', 's7925966g', 's7926969g', 's7927961g', 's7928964g',
's7929967g', 's7931962g', 's7932965g', 's7933968g', 's7934960g',
's7935963g', 's7936966g', 's7937969g', 's7938961g', 's7939964g',
's7940967g', 's7942962g', 's7943965g', 's7944968g', 's7945960g',
's7946963g', 's7947966g', 's7948969g', 's7949961g', 's7950964g',
's7951967g', 's7953962g', 's7954965g', 's7955968g', 's7956960g',
's7957963g', 's7958966g', 's7959969g', 's7960961g', 's7961964g',
's7962967g', 's7964962g', 's7965965g', 's7966968g', 's7967960g',
's7968963g', 's7969966g', 's7970969g', 's7971961g', 's7972964g',
's7973967g', 's7975962g', 's7976965g', 's7977968g', 's7978960g',
's7979963g', 's7980966g', 's7981969g', 's7982961g', 's7983964g',
's7984967g', 's7986962g', 's7987965g', 's7988968g', 's7989960g',
's7990963g', 's7991966g', 's7992969g', 's7993961g', 's7994964g',
's7995967g', 's7997962g', 's7998965g', 's7999968g']

Enter NRIC with 1-4 masked digits (<Enter> to exit):
Program exited
```

(20 marks)

Question 4 (25 marks)

Read the entire question before attempting. You need to submit each part individually.

A simplified "Treasure Hunt" game can be implemented using a list.

For example, in the following list represents 10 slots, containing SIX '1's which are the treasures, TWO '0's representing empty slots and TWO '-1's as "bomb". The first slot is index 0, so if the player enters "0,1,2,3", the score is the sum of the first 4 slots which is 2 (0+1+0+1). If the player enters "3,4,5,6"as picks, the game will return "You hit a bomb!!" as index 4 contains "-1", and the game will not reveal any score.

[0, 1, 0, 1, -1, 1, 1, -1, 1, 1]

The program generates this hidden list, and players will have 5 chances to find 4 treasures.

The game rules are as follows:

1. At start of the game, create a new list of length 10, with SIX '1's, TWO '0's and TWO '-1's randomly placed into the list.

2. Obtain the 4 picks (indices) from player. Ensure the picks are valid indices (0-9).
3. Determine the score, taking into account the bomb, if any.
4. If FOUR treasures are located, display a message that the player has won and the game ends.
5. If not, repeat from step 2 until player exhausted all 5 chances.
6. Display message that player has lost and show where the treasures are hidden.

<u>Sample run 1:</u> The treasures are planted... You have 5 chances!! Round 1: Enter your 4 picks: **1,3,5,7** You hit a bomb!! Round 2: Enter your 4 picks: **1,4,5,7** You got 3 hits!! Round 3: Enter your 4 picks: **1,4,7,6** You got 4 hits!! You are the winner!!	**Explanation:** The hidden list is [1, 1, 0, -1, 1, 0, 1, 1, -1, 1] Index 3 is "-1" 3 treasures located 4 treasures located
<u>Sample run 2:</u> The treasures are planted... You have 5 chances!! Round 1: Enter your 4 picks: **2,4,6,8** You hit a bomb!! Round 2: Enter your 4 picks: **3,4,6,8** You hit a bomb!! Round 3: Enter your 4 picks: **0,3,4,8** You got 3 hits!! Round 4: Enter your 4 picks: **0,3,4,9** You got 3 hits!! Round 5: Enter your 4 picks: **3,4,8,9** You got 3 hits!! You lost!! The treasures are in: [1, -1, 1, 0, 1, 0, -1, 1, 1, 1]	The hidden list is [1, -1, 1, 0, 1, 0, -1, 1, 1, 1] Index 6 is "-1" Index 6 is "-1" 3 treasures located 3 treasures located 3 treasures located

a) The following 2 functions are the building blocks for the "Treasure Hunt" game:

```
getTreasureList( ) -> list:
```

- This function returns a newly shuffled list of integers, containing SIX '1's, TWO '0's and TWO '-1's randomly placed.
- For example: [0, -1, 0, -1, 1, 1, 1, 1, 1, 1]

```
countTreasures(treasureList: list, indices: list) -> int:
```

- Given the parameter indices (a list of 4 integers), the function will use these 4 indices to locate and count the treasures found in the parameter treasureList.
- If any of the 4 indices located the "bomb", the function returns -1.

Write the `getTreasureList( )` and the `countTreasures( )` functions.

(10 marks)

b) Using the 2 functions created in part (a), write the `main( )` function that allows a single player to play the simplified "Treasure Hunt" game.

(15 marks)

---- END OF ASSIGNMENT ----